1. Temperature and thermal equilibrium

Temperature describes how hot or cold an object is and determines the direction of heat transfer. When objects at different temperatures are in thermal contact, energy flows from the hotter one to the colder one until thermal equilibrium is reached. Equal temperatures do not imply equal internal energies: material, mass and physical state also matter.

2. Fixed points of the Celsius scale

The ice point is the temperature of melting pure ice at standard atmospheric pressure, defined as 0 °C. The steam point is the temperature of steam above boiling pure water under the same pressure, defined as 100 °C. Calibration requires the specified pressure, pure substances and steady readings. The steam point is measured in the steam.